

Does wavefunction collapse require a conscious observer? A preregistered retrocausal interferometer test with human and artificial intelligence observers

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Abstract

The von Neumann–Wigner framework holds that wavefunction collapse is incomplete until a conscious observer intervenes. Wheeler argued that measurement choices can be deferred until after a photon transits the apparatus. Together, these ideas raise fundamental questions: What is an observer, and when does observation become definite? To investigate, diffraction-grating data recorded months before human observation drove an online animated candle flame across 1,000 human and 1,000 AI-agent sessions. Sessions alternated ten 30-second intention epochs (intending flame dimming) and ten relax epochs. One first-order diffraction spot drove the display; the other served as an unobserved control. Participants were blind to the control and prerecording; assignments were randomized. The observed sensor was regressed against the unobserved sensor, and slopes compared via label-swapping permutations. Of eight possible outcomes, only the human-observed-intention cell was significant after multiple-comparison correction ($p = 0.00007$). AI agents showed no effect. Pre- and post-preregistration windows were statistically equivalent. These results suggest an effect of human consciousness rather than hardware drift or analytical artifacts.

Keywords: Quantum measurement problem; Von Neumann-Wigner interpretation; wavefunction collapse; retrocausation; consciousness; observer effect; AI agent; preregistered; delayed-choice; optical interferometer

1 Introduction

1.1 The measurement problem

The question whether consciousness plays a role in collapse of the quantum wavefunction is among the oldest unresolved problems in the foundations of physics. In the von Neumann–Wigner framework, physical systems evolve continuously and deterministically under the Schrödinger equation but discontinuously and irreversibly when a measurement is performed [1,2]. The theory itself is silent on what constitutes a measurement, and von Neumann argued that the “cut” between quantum and classical descriptions can be placed anywhere along the chain, from observed system to apparatus to nervous system to mind, without altering empirical predictions. London and Bauer located the cut at the moment a conscious observer becomes aware of an outcome [3]. Wigner elevated this into a positive thesis by proposing that the equations of quantum mechanics cease to be linear when conscious beings enter the picture, and the mind itself selects a definite branch of the superposition [4]. Wigner later expressed reservations about this position, partly in response to the Wigner's-friend solipsism problem and also out of growing sympathy with objective-collapse alternatives [5]. Thus, we use “von Neumann–Wigner” (vN-W) in its now-standard sense as a label for an interpretive position, rather than as a claim about Wigner's later position.

Three broad objections have been raised against the vN-W interpretation. The conceptual objection is that introducing consciousness as a physical agent appears to violate parsimony or it leads to solipsism [6–8]. The biophysical objection is that the macroscopic apparatus through which any photon must pass, i.e. cornea, retinal layers, photoreceptor isomerization, ganglion spike trains, etc., provides ample opportunity for objective state reduction long before any signal reaches awareness [9]. The interpretive objection, developed by French among others, holds that the London-Bauer analysis is better read as concerning the correlation between observer and observed within a Husserlian framework, rather than as the physical collapse mechanism Wigner later attributed to it [3,10].

Despite such objections, several quantitative consciousness-collapse models have been developed, including Stapp's quantum-Zeno-effect proposal in which conscious attention extends the lifetime of selected neural states [11]; the Chalmers-McQueen integration of

integrated information theory with continuous spontaneous localization, which produces explicit empirical predictions [12], and related quantitative proposals from Kent [13], Kremnizer and Ranchin [14], and Okon and Sebastián [15]. Several recent quantitative models have sharpened the hypothesis into testable form [16]. The present design addresses this concern directly. If human and AI agents produce statistically equivalent results on the observed sensor, then the vN-W prediction would be disconfirmed.

In addition, the foundations literature has slowly returned the observer to serious consideration. Frauchiger and Renner showed in 2018 that contradictions arise when quantum theory is applied consistently to observers who themselves contain observers [17]. A Nature survey of more than 1,100 physicists in 2025 found that nine percent of respondents held that a measurement specifically requires a conscious observer, about the same order of magnitude as support for Bohmian mechanics [18]. Nicolas Gisin recently emphasized the stakes: “If someone does the [consciousness-collapse] experiment and gets a surprising result, the reward is enormous. It would be the first time we as scientists can put our hands on this ... problem of consciousness” [19].

1. 2 Empirical literature

Theoretical discussions and empirical tests of the consciousness-collapse question have been reported for over a half-century [20,21]. Beloff and Evans (1961) asked if observers could influence radioactive decay from a sodium iodide source [22]. Hall, Kim, McElroy, and Shimony (1977) reasoned that if person A’s observation collapsed an event’s wavefunction, then person B might be able to subjectively detect that collapse. Their main study of 544 trials gave results exactly at chance, although a similar study they mentioned, using cobalt-57, yielded 40 hits in 67 trials ($p < 0.07$) [23].

A more systematic empirical program using devices based on quantum randomness was launched by Schmidt in 1970 and later pursued by many researchers [24], including a multi-decade research program at Princeton University [25]. A meta-analysis of those studies reported a small magnitude effect at 6.8σ beyond chance ($p < 10^{-11}$) with no evidence of selective reporting or obvious methodological flaws [26]. A later meta-analysis on a subset confirmed a small effect at 4.1σ ($p < 2.1 \times 10^{-5}$) but flagged three large-scale studies showing deviations

opposite to the direction of intention [27]. The authors attributed the results to publication bias, but independent reanalyses argued that this could not explain the outcomes [28,29].

More recent experiments have explicitly tested the vN-W proposal using optical interferometers [30–37]. These studies asked participants to mentally obtain which-path information about photons in the apparatus, or to intentionally influence the path that the photons travelled. Across roughly 30 experiments at six labs, 14 tests were reportedly statistically significant (two-tailed $p < 0.05$), giving a cumulative binomial probability below 10^{-10} . Significance counting is only suggestive, but an independent review concluded that collectively these experiments seemed to provide positive evidence of a weak interaction between consciousness and quantum systems [38], and several other experiments using different designs have shown similar effects [39,40]. Vigorous debates over interpretations and methods have ensued [41–43], and in parallel a theoretical literature has developed mathematical formalisms within orthodox quantum mechanics that propose to account for observation-dependent retrospective correlations without requiring departures from standard quantum theory [44–46].

1.3 The present study

The present experiment extended this line of research in three ways. First, it used a diffraction grating interferometer instead of a double-slit arrangement, providing a more straightforward way to analyze the hypothesized observer effect than methods required by double-slit interference patterns. Second, it incorporated an AI-agent comparison arm run on identical hardware, software, and analysis pipelines, providing a within-apparatus control that directly tested if whatever produces a purported human observer effect is specific to human conscious experience. Third, the retrocausal design, using prerecorded sensor data to drive the participant's interface to the optical apparatus, eliminated by construction the possibility that classical environmental coupling between participant and apparatus could explain the results. Together these features sharpened the empirical question about the consciousness-collapse concept.

2 Materials and Methods

2.1 Hardware

The optical apparatus consisted of an ultra-low-noise diode laser (Coherent ULN module, 635 nm, 5 mW, circular beam, RMS noise < 0.06% from 10 Hz to 10 MHz) directed through a transmission diffraction grating (Thorlabs GT25-06V, 600 grooves/mm, 28.7° groove angle, 25 × 25 mm) mounted on a Thorlabs optical breadboard. Two Thorlabs PM100D power meters were positioned at the +1 and −1 first-order diffraction spots. These components were arranged as shown in Figure 1.

The blazed grating on the transmission grating meant it had asymmetric groove faces that concentrated diffraction efficiency into one particular order at the expense of the other. This asymmetry resulted in a ~4:1 power ratio (2.4 mW vs 0.6 mW) in the two sensors. This asymmetry was selected to provide two distinct photon-flux conditions within the same apparatus.



Figure 1. Implementation of the optical diffraction system.

We computed signal-to-noise ratio directly from 10 representative raw experimental recordings (total of 10 hours). The lower-power diffraction spot (mean 0.608 mW) yielded $\text{SNR} = 65.8 \text{ dB}$ (range 59–73 dB across files), with RMS noise of $0.34 \mu\text{W}$ (0.056% of mean power). The higher-power spot (mean 2.423 mW) yielded $\text{SNR} = 72.3 \text{ dB}$ (range 64–75 dB), with RMS noise of $0.63 \mu\text{W}$ (0.026% of mean power). Both figures are consistent with or better than the manufacturer's specification of $<0.06\%$ RMS over 10 Hz–10 MHz. The low-frequency (0–1 Hz) band sampled at 2 Hz did not appear to add substantial $1/f$ noise beyond published specifications. These measured SNR values confirm that classical intensity noise, while the dominant noise source, did not preclude detection of systematic slope differences via the regression-residual method.

2.2 Software

Optical power and detector temperature were recorded by the Thorlabs Optical Power Monitor application (v6.1.4945.765_NSIS) at 2 Hz. The web app used to run the experiment online was implemented in React.js (framework), Tailwind CSS (for page styling), and Redux toolkit (for API Integration). The mobile versions were implemented using React Native (a mobile app cross-platform framework), Redux Toolkit (for API integration), Async Storage (for session storage), and React Native Web view (for the animations). The backend was Express.js (Nodejs Framework), MongoDB (database), BullMQ (for managing Jobs and Queues), Redis (memory cache), AWS Elastic Beanstalk, and an AWS S3 bucket (for images and audio). Deployment for the web app was Netlify, Google Play Console for the Android app, and the Apple App Store for the iOS app.

All data analyses were written in Matlab R2025b with assistance by Claude Code (Opus 4.7). The data and Python-based analytical scripts are available on Zenodo.com, and the analyses can be run on Google Colab (see details in the Data Availability section).

2.3 Sensor data preprocessing

Optical power data were recorded continuously at 2 Hz. To allow for thermal and mechanical stabilization, the first 50 minutes of each recording session were discarded. The remaining data were segmented into sequential 1,200-sample (10-minute) arrays. The higher-power channel (~2.4 mW) was assigned to an “A” array and the lower-power channel (~0.6 mW) to a “B” array.

Each 10-minute array was then detrended and z-score normalized to remove slow thermal drift. Processed segments were stored as column pairs labeled as A1/B1, A2/B2, The observed/unobserved sensor assignment alternated across pairs (i.e., A1, B2, A3, B4, ...), so neither physical sensor was systematically privileged.

None of the raw or processed sensor data were observed by the experimenters before any participant ran a session, and participants were unaware that the experiment used a retrocausal design or that the data driving the candle animation alternated between sensors across sessions.

Raw sensor data were recorded from June 7, 2025 to April 13, 2026 in a total of 73 recording sessions ranging from about 50 minutes to 18 hours, with a median of 6.6 hours.

2.4 Online experiment design

User interface. The user interface resembled a simple adventure game accessible via a web browser or an iPhone, iPad, or Android device. In twenty alternating epochs of 30 seconds each, users were asked to view a dynamically animated candle flame (see Figure 2), while mentally intending that the flame become dimmer (intention assignment), or to withdraw their attention when the flame was uniformly dim and static (relax assignment). The assignment to “intend” was used to actively support the user’s sustained attention. By comparison, a more passive instruction, such as “gaze at the flame,” would have invited mind-wandering [47,48].

During each intention epoch, a cloud-like shape next to the flame resolved into a discernible word (selected at random out of a pool of 500 pre-set words) when the flame dimmed. An ambient soundscape volume also varied with the flame’s brightness.^a During the following relax epoch, the user was asked to select that same word out of four alternatives. This served as an attentional engagement task, whereby at least 9 of 10 words had to be correctly identified to be included in the formal analysis.

^a *Healing Sounds: Frequencies II*, “Reiki Chants,” by Jonathan Goldman, healingsounds.com

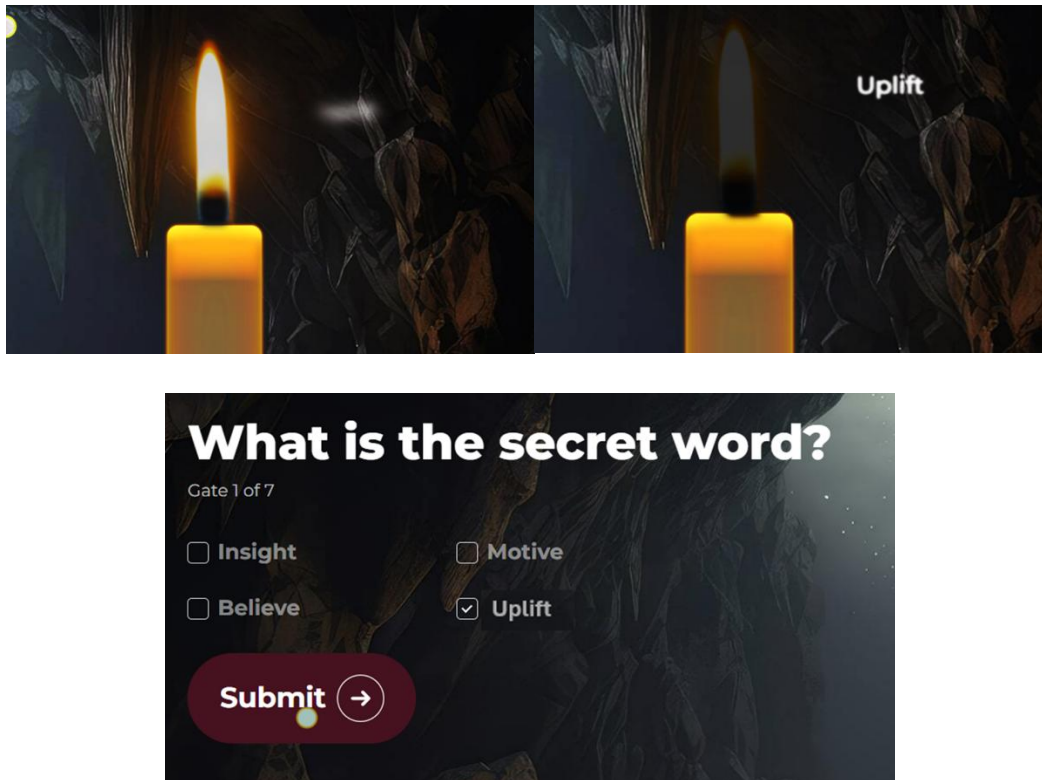


Figure 2. The upper left image is a snapshot of a moment during an “intention” epoch, with a bright animated candle flame near a cloud-like shape. The upper right shows the candle when the flame dims, whereupon the adjacent cloud coalesces into a more easily visible word. The bottom panel shows the screen in the following “relax” epoch, where the user is asked to select the word that they saw.

The flame brightness, soundscape volume, and word visibility were all driven by prerecorded data from the observed first-order spot. The unobserved spot's data played no role in the participant's experience. Note that the terms observed and unobserved refer to the sensor data played back during a session, whereas intention and relax refer to the attentional assignments within a session.

To encourage repeated participation, a leaderboard displayed the session score, which was the sum of data array values during the intention epochs minus the sum in the relax epochs, with the most negative sum at the top (because the task was to make the flame dimmer). After a user successfully completed the first session with at least 9 out of 10 words correctly identified, an option was given to proceed to another “gate.” The experiment consisted of seven gates, each

with different background imagery and flame styles, but the underlying experimental design was identical in each case. The leaderboard displayed the cumulative score of all valid trials across all gates contributed by each user.

Questionnaire and consent. At registration, participants agreed to an informed consent and completed a questionnaire covering demographics, education, Big Five personality [49], meditation experience, interest in traditional magical practices, self-rated creativity, insight experience, and a 10-item paranormal belief scale. Then they started a test session. The protocol was approved by the Institute of Noetic Sciences Institutional Review Board (document RADD_226_01).

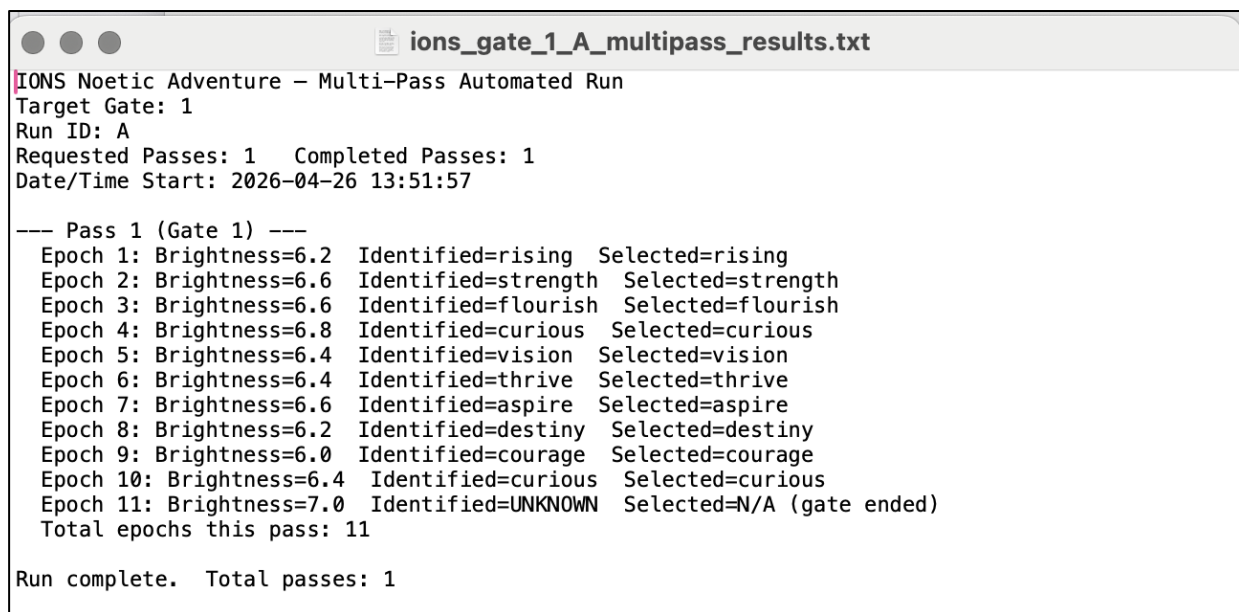
Website backend. Investigators uploaded a datafile containing arrays of the preprocessed sensor data to the web host server, in the form described above (e.g., a matrix 1200×1500 , where each row contained sensor sample data, and each column corresponded to a unique session). When a user started a session, the web program selected the next unused array and used it to drive the candle animation at 2 Hz, the same rate as the original data recording. With 1200 samples per array, a single run took 10 minutes.

Data downloaded from the website consisted of one row per user, with columns containing user IDs, anonymizing codes, optionally contributed email addresses, when each session began, what “gate” number was run, the score for that run, the ten words presented in the session, how long it took the user to select the word, and whether it was correct. The preregistered analysis specified that only completed sessions with 9 of the 10 words correctly answered were part of the formal database.

Data integrity. Because the data arrays used to drive each session were recorded (but unobserved) months before playback, and each array was identified by a code number, all analyses used the array code to refer to the original data array, which was stored offline. Checks of per-session scores determined by the web app detected no anomalies suggestive of attempts to tamper with the online code or database.

2.5 AI observer

An AI observer was implemented via Claude Code (Opus 4.6, version: 2.1.74) to simulate a human session at the browser level. The agent (a) viewed the same animated candle a human would see, sampling flame brightness on a 0-10 scale using a Claude vision API (Opus 4.6) five times per intention epoch within a defined region of interest, (b) read the cloudy word during the intention epoch via a separate vision-API call (Haiku 4.5), and (c) selected the matching word during the following relax epoch using Playwright for Python with Firefox. The same 9-of-10 word criterion used to qualify human sessions was applied to the AI sessions. After each session a report file was written listing recorded brightness values and word identifications per epoch (see Figure 3).



```
ions_gate_1_A_multipass_results.txt
IONS Noetic Adventure – Multi-Pass Automated Run
Target Gate: 1
Run ID: A
Requested Passes: 1   Completed Passes: 1
Date/Time Start: 2026-04-26 13:51:57

--- Pass 1 (Gate 1) ---
Epoch 1: Brightness=6.2   Identified=rising   Selected=rising
Epoch 2: Brightness=6.6   Identified=strength Selected=strength
Epoch 3: Brightness=6.6   Identified=flourish Selected=flourish
Epoch 4: Brightness=6.8   Identified=curious  Selected=curious
Epoch 5: Brightness=6.4   Identified=vision   Selected=vision
Epoch 6: Brightness=6.4   Identified=thrive   Selected=thrive
Epoch 7: Brightness=6.6   Identified=aspire   Selected=aspire
Epoch 8: Brightness=6.2   Identified=destiny  Selected=destiny
Epoch 9: Brightness=6.0   Identified=courage  Selected=courage
Epoch 10: Brightness=6.4  Identified=curious  Selected=curious
Epoch 11: Brightness=7.0  Identified=UNKNOWN  Selected=N/A (gate ended)
Total epochs this pass: 11

Run complete. Total passes: 1
```

Figure 3. Example report written by an AI agent upon completing a session.

The AI provided two controls: It tested if information extraction by a (presumably) non-conscious system would reproduce results similar to humans, and it also provided a way to independently vet the online display and analysis pipeline, which were identical for human and AI sessions, to test if those methods generated spurious results.

The AI agent read the secret word via repeated vision-API calls throughout the intention epoch, submitting a word-identification call for each screenshot frame and selecting the plurality candidate once three or more confident votes accumulated, which was typically within the first several seconds of the epoch. Word reading was not conditioned on any observed brightness trend; the brightness assessments (five samples at fixed intervals) and word-identification calls ran in parallel threads with no dependency between them.

2.6 Primary hypothesis

The metric of interest was the linear slope of illumination at each sensor during the ensemble means of the intention and relax epochs, based on a gradual-decline pattern reported in a prior experiment [43]. Four slopes were computed for each cohort (human, AI), yielding eight test cells. The consciousness-collapse hypothesis was that the human $\times$ observed $\times$ intention cell would be significantly negative. The other seven cells were predicted to be null.

A negative slope during the intention epochs would mean that the z-scored optical power at the observed sensor decreased progressively over the 30-second epoch window, consistent with the hypothesis that sustained conscious intention gradually reduced the first-order diffraction intensity, as predicted by the vN-W wavefunction-collapse hypothesis. A flat or positive slope during relax epochs served as a within-session control. If the effect was specific to intentional engagement, then the relax slopes should not drift downward.

2.7 Preregistered analysis

Fixed analysis parameters were established based on January–March 2026 pilot data, including (1) a session-qualification criterion of at least 9 of 10 words correctly identified, (2) a 5-second delay due to the cognitive cost of attention-switching between the intention and relax assignments [50] (based on observations in similar experiments [28,31]), and (3) all sessions completed across all gates (see <https://osf.io/6qutr/registrations>).

The preregistration was prospective with respect to the stopping rule (1000 qualified sessions by humans and AI) and to the final analysis criteria, but not with respect to all human data collection. Therefore, the result should be regarded as a preregistered analysis of a partly pre-existing dataset, with a pre/post equivalence analysis serving as a safeguard against investigator

analytical tuning. We also note that the AI sample size was increased from 500 to 1000 after preregistration to match the statistical power afforded by the 1000 session human trials.

Because both sensors measured light from the same laser, they shared a common noise component. To isolate observation-specific signal, the observed sensor was regressed on the unobserved sensor ($\text{observed} = b_0 + b_1 \times \text{unobserved} + \text{residual}$). The residual represented observed-channel activity not explained by shared laser fluctuations. Each session was then divided into ten alternating one-minute blocks, where each block comprised a 30 s intention epoch followed by a 30 s relax epoch. Within each block, the first 5 s of each half were discarded to account for cognitive lag, leaving 25 s per epoch. A linear slope was fitted to each 25 second segment, yielding $1000 \text{ sessions} \times 10 \text{ epochs/session} = 10,000$ paired slope measurements for intention epochs, and the same for relax epochs.

The preregistered analysis asked if the intention-relax slope difference was significantly negative. To accomplish this, the intention/relax labels were swapped within blocks across 5,000 permutations to build a chance distribution. The actual difference was then compared to the null distribution to yield a one-tailed p-value. Operating on block-level slopes rather than raw samples made the test robust to within-epoch autocorrelations. The same battery of tests was applied to the unobserved sensor as a built-in control.

3 Results

3.1 Participants

Some 295 unique individuals contributed to the 1,000 qualifying human sessions. Mean age was 37.0 (sd 13.0, range 14–63); 161 were male, 127 female, 6 “other”, and 1 missing entry. Education ranged from high school through doctoral degrees; 92.5% reported at least one experience of insight; 69.2% had a regular meditation practice [mean 13.8 years]; 19.3% reported interests with traditional magical practices; and mean self-rated creativity was 79.5 of 100. Most participants were from the United States ($N = 182$) and Canada ($N = 15$).

None of the paranormal belief, demographic, or practice variables predicted performance after correction for multiple comparisons. The only nominally significant predictor was the Big Five trait of openness ($r = -0.119$, $p = 0.043$), where a negative correlation corresponded to better

task performance (the observed spot slope declining). As the only interesting result in an exploratory panel of 26 candidate variables, this outcome is suggestive pending replication.

3.2 Qualified sessions

As of April 25, 2026, the number of sessions recorded and included were as follows in Table 1. The original target for AI sessions was 500 sessions (powered at 80% to detect the human effect size of Cohen's $d = 0.035$). We subsequently raised that target to 1,000 AI sessions to match the human cohort and provide equal statistical power for the comparison. The stopping rule remained chronological, the first 1,000 qualifying sessions by date.

For the qualifying human sessions, correct responses to select the word during the relax epochs were faster than incorrect responses, with means of 34.4 s vs. 40.7 s respectively, a mean difference of 6.3 seconds (Welch $t(169) = 12.38$, $p < 10^{-25}$, Cohen's $d = 2.36$). Given that word-selection options appeared after the 30 second intention epoch, these times correspond to decision latencies of ~ 4 s for correct responses and ~ 10 s for incorrect responses, a pattern consistent with genuine effortful memory retrieval rather than arbitrary clicking.

Table 1. Qualified sessions required to achieve the preregistered 1000 sessions each for human and AI observers, as of April 25, 2026.

	Human	AI
Total sessions recorded	1,543	1,233
Excluded: fewer than 10 epochs completed	152	17
Excluded: fewer than 9/10 words correct	203	142
Qualifying sessions	1,188	1,074
Excluded: collected after preregistered target reached	188	74
Analyzed	1,000	1,000

3.3 Preregistered tests

The preregistered regression-residual analysis was applied to the first 1,000 qualifying human sessions (recorded January 31 to April 25, 2026) and the first 1,000 qualifying AI sessions (recorded March 20 to April 22, 2026). Three permutation tests (5,000 iterations each) were applied to the regression residual: (a) for the observed sensor the intention, the slope vs. zero was calculated, (b) the relax slope vs. zero, and (c) the intention-relax difference via within-block label swap. The same tests were applied to the unobserved sensor.

Human cohort. At the preregistered lag of 5 s, the residual intention vs. relax slope was -4.13×10^{-4} , $p = 0.0002$, indicating a reliable decline at the observed sensor during intention epochs, confirming prediction 1 (see §S2 for other lags). The residual relax slope vs. zero was $+1.13 \times 10^{-4}$, $p = 0.830$, consistent with the expected null. The intention-relax difference was -5.27×10^{-4} , $p = 0.0016$ in the per-block test and $p = 0.0014$ in the ensemble test (Cohen's $d = -0.030$). Figure 4 shows the intention and relax ensemble slopes separately, along with comparison to the permuted null distributions.

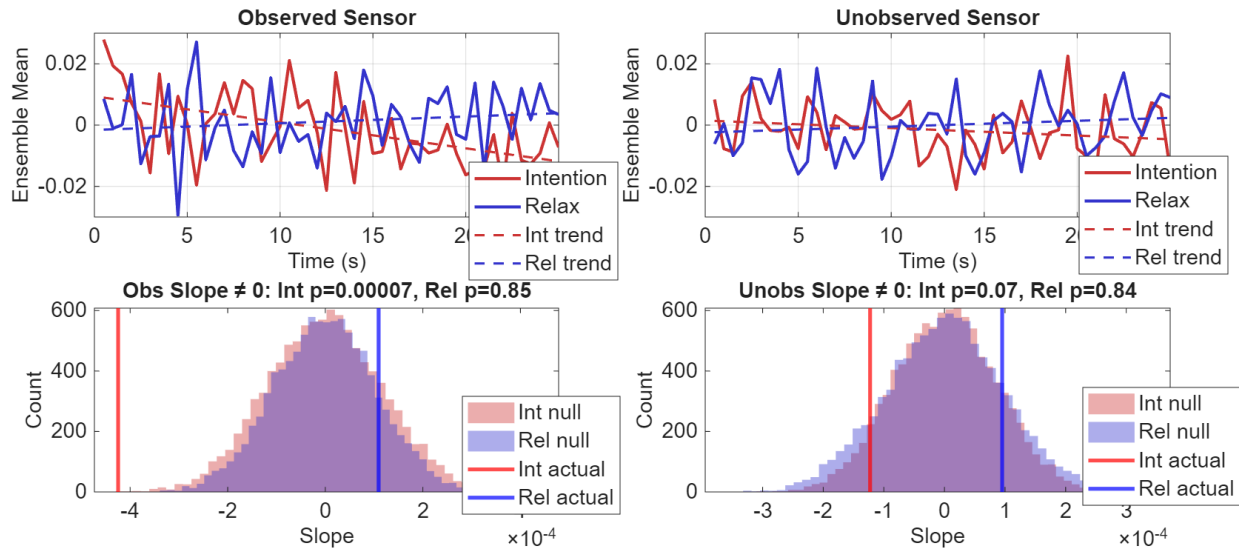


Figure 4. Humans as observers. (Top) Intention (Int) and relax (Rel) ensemble slopes for observed and unobserved sensors. (Bottom) Intention and relax slopes compared to permuted null distributions.

The per-block test and ensemble test addressed complementary questions. Because block denoted one intention/relax pair, the per-block test treated each block's slope difference as the unit of analysis and asked if the distribution of all 10,000 such differences departed from a within-block label-swap null. This was sensitive to whether the effect appeared consistently at the block level. By comparison, the ensemble test averaged across every block and session into a grand-mean intention curve and a grand-mean relax curve; it asked if the slope of the resulting 50-sample aggregate (the intention and relax epochs were each 25 samples) departed from a time-shuffled null, and as such it was sensitive to a coherent linear trend in the averaged time course. These two tests can disagree in principle, but concordance across both, as observed here, provided stronger evidence than either one alone. The parametric ensemble fit yielded slope $p = 1.3 \times 10^{-4}$, $R^2 = 0.246$.

For the unobserved sensor the intention slope was -1.23×10^{-4} , $p = 0.069$; the relax slope was $+9.5 \times 10^{-5}$, $p = 0.825$; and the intention-relax difference was -2.18×10^{-4} (per-block $p = 0.113$, ensemble $p = 0.110$). Prediction 2 was therefore satisfied at the $\alpha = 0.05$ threshold. The borderline significant intention slope is addressed in the Discussion section.

AI cohort. The residual intention slope was -2.3×10^{-5} , $p = 0.501$; the residual relax slope was -1.50×10^{-4} , $p = 0.083$; the intention-relax differential was $+1.27 \times 10^{-4}$ (per-block p

= 0.785, ensemble $p = 0.790$, Cohen's $d = +0.008$). The unobserved-sensor tests were equally null (intention slope $p = 0.325$, relax slope $p = 0.287$, differential per-block $p = 0.543$, $d = +0.001$). Prediction 3 was thus confirmed: AI agents produced no detectable signal on any preregistered measure. Figure 5 shows the AI agent intention and relax ensemble slopes separately, along with comparison to the permuted null distributions.

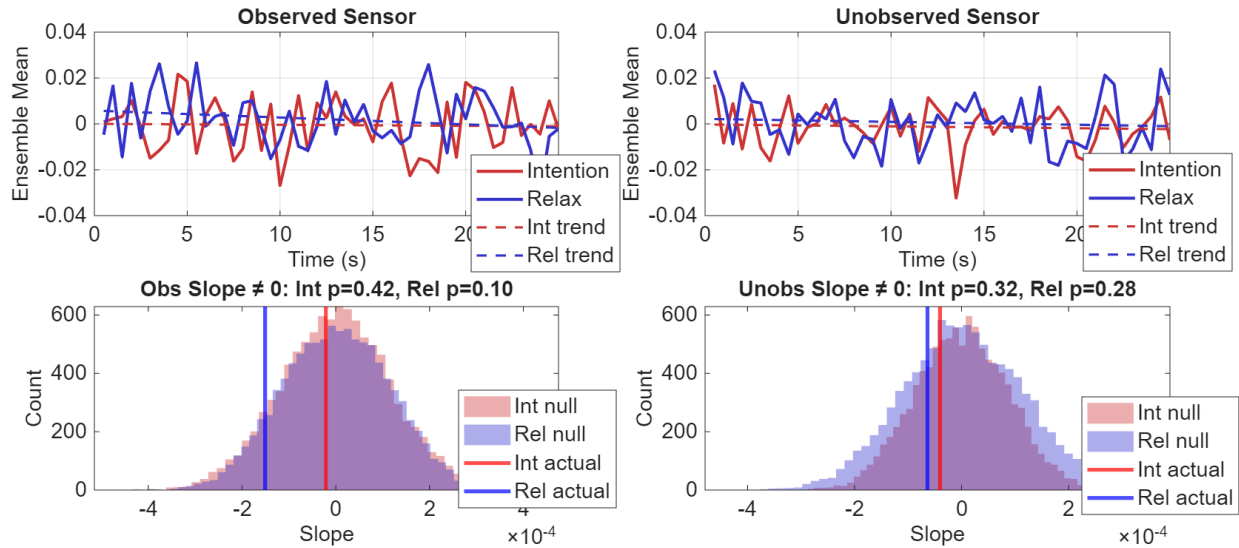


Figure 5. AI agent as observer. (Top) intention and relax ensemble slopes for observed and unobserved sensors. (Bottom) Null distributions in relationship to the intention and relax slopes.

3.4 Exploratory power-level dependence

Sessions were stratified by the power level of the observed sensor, yielding 493 sessions with the high-power channel observed, and 507 sessions with the low-power channel observed. The intention-relax slope difference for the high-power channel was non-significant: intention slope = -2.53×10^{-4} , relax slope = -2.3×10^{-5} , paired $t(4929) = -1.02$, $p = 0.307$, Cohen's $d = -0.015$. Permutation tests on the ensemble curves confirmed the null, with intention slope $p = 0.123$, relax slope $p = 0.895$. When the low-power channel was observed, the intention-relax slope difference was significant, with the intention slope = -5.91×10^{-4} , relax slope = $+2.38 \times 10^{-4}$, paired $t(5069) = -3.04$, $p = 0.002$, Cohen's $d = -0.043$. Permutation tests on the ensemble curves gave intention slope $p = 0.0002$, relax slope $p = 0.095$.

The high-power arm carried $\sim 4\times$ the mean optical power and $\sim 4.1\times$ the detrended RMS noise of the low-power arm. The signal to noise ratio was therefore essentially equal across the two sensors, indicating that detection of an observer effect was limited by amplitude noise rather than shot noise. The intention–relax slope difference was $3.6\times$ larger in absolute magnitude on the high-power arm, but its fractional magnitude (per unit mean power) and its effect-to-noise ratio were both $\sim 1.4\times$ larger on the low-power arm. The theoretical implications of this observation, namely that fewer photons per measurement interval produced a larger effect is developed in the Discussion.

3.5 Factorial summary

The eight slope-against-zero tests can be displayed as a single $2 \times 2 \times 2$ factorial, crossing observer (human, AI), sensor (observed, unobserved), and epoch (intention, relax) (see Figure 6 and Table 2). The consciousness-collapse hypothesis predicted significance only in the human $\times$ observed $\times$ intention cell, and of the eight results only that cell was significant after applying Benjamini–Hochberg false-discovery-rate (FDR) correction [51]. The human $\times$ unobserved $\times$ intention cell trended at $p = 0.074$, with a slope of -0.00012 , as compared to the observed-sensor magnitude of -0.00042 .

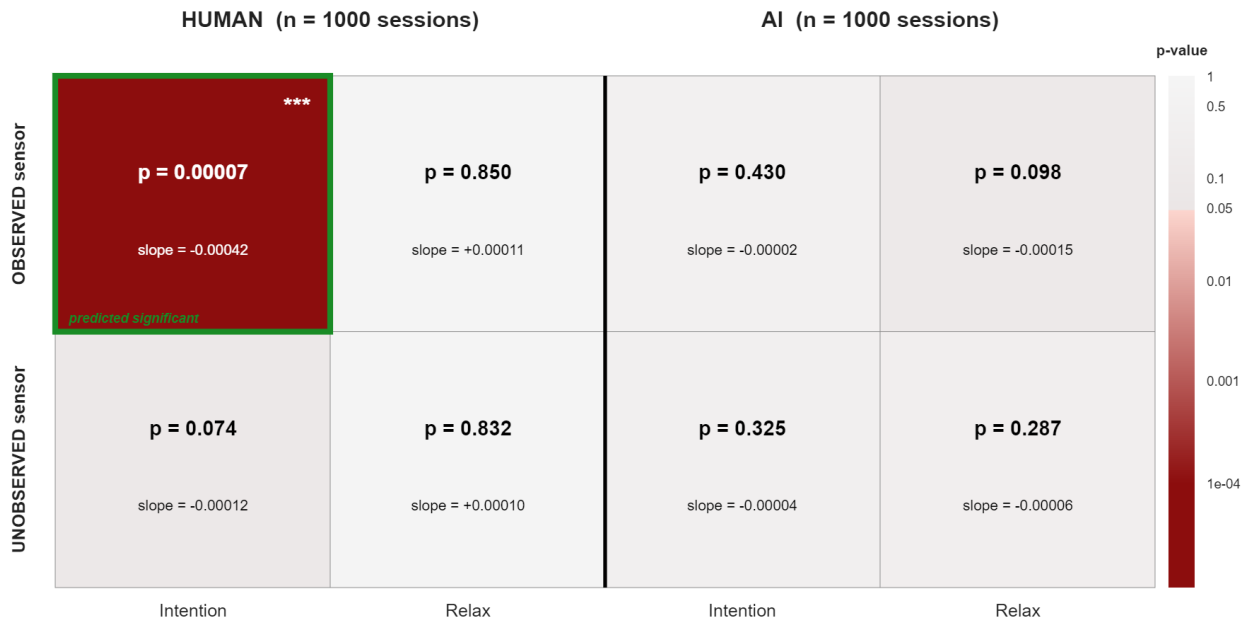


Figure 6. Eight-cell summary of the ensemble mean epoch slopes. The observed $\times$ human $\times$ intention cell was predicted to be significant, the other cells were expected to produce null results. The p-values shown in the cells are parametric p-values uncorrected for multiple comparisons.

Table 2. Complete 2x2x2 table with confidence intervals.

Agent	Sensor	Epoch	Slope	95% CI	z	p (one-tailed)
Human	Observed	Intention	-0.000425	[-0.000656, -0.000194]	-3.57	0.00007
Human	Observed	Relax	+0.000109	[-0.000097, +0.000315]	+1.05	0.851
Human	Unobserved	Intention	-0.000123	[-0.000290, +0.000044]	-1.46	0.074
Human	Unobserved	Relax	+0.000095	[-0.000097, +0.000287]	+0.97	0.832
AI	Observed	Intention	-0.000021	[-0.000252, +0.000210]	-0.18	0.430
AI	Observed	Relax	-0.000151	[-0.000376, +0.000074]	-1.32	0.098
AI	Unobserved	Intention	-0.000040	[-0.000210, +0.000130]	-0.47	0.325
AI	Unobserved	Relax	-0.000064	[-0.000283, +0.000155]	-0.58	0.287

3.6 Pre/post-preregistration equivalence

Because the preregistered analysis plan was filed after a portion of the human data was collected, we tested if the pre-preregistration window (January–March 2026) and the post-preregistration window yielded equivalent estimates (April–May 2026, with an extended exploratory window of $N = 308$ sessions used for the out-of-sample validation; see §S4 for details). The pre-window intention-relax slope was -5.33×10^{-4} ($d = -0.029$, 95% bootstrap CI $[-0.052, -0.008]$), and the post-window value was -5.37×10^{-4} ($d = -0.035$, 95% CI $[-0.079, +0.011]$). Welch's t on the between-window difference was $t(3096) = 0.008$, $p = 0.994$; a 2,000-iteration permutation test resulted in $p = 0.992$.

Five additional replication criteria drawn from the open-science literature were also satisfied (see Table 3) [52–57]. They showed the same direction of effect, bootstrap confidence-interval overlap, post-window mean within the pre-window prediction interval, small telescopes effect ($|d_{\text{post}}| \geq 33\%$ of $|d_{\text{pre}}|$), and two one-sided tests (TOST) equivalence within a ± 0.10 d-unit margin. The two time windows were statistically indistinguishable on the primary statistic, and therefore the inclusion of pre-preregistration sessions in the confirmatory analysis did not indicate over-fitting. See §S1 for another way to demonstrate cumulative consistency in effect size.

Table 3. Replication criteria comparing pre-preregistered data and post-preregistered data.

Criterion	Result	Evidence
Same direction of effect	PASS	Pre: $d = -0.029$ Post: $d = -0.035$
Small telescopes ($ d_{\text{post}} \geq 33\%$ of $ d_{\text{pre}} $)	PASS	$ d_{\text{post}} = 0.035$ vs. threshold 0.010 (121% of pre)
Bootstrap CI overlap on Cohen's d	PASS	Pre CI $[-0.052, -0.008]$ Post CI $[-0.079, +0.011]$
Post mean within prediction interval	PASS	PI $[-0.001458, +0.000391]$ observed $= -0.000537$
TOST equivalence (± 0.10 d-units)	PASS	$p < 0.0001$

4 Discussion

4.1 Summary of findings

Of the eight cells in the observer $\times$ sensor $\times$ epoch factorial, the only cell significant after correction for multiple comparisons was the one predicted by the vN-W interpretation. Several interpretations of quantum mechanics predict no such asymmetry. In relational quantum mechanics [58], many-worlds [59], and decoherence-based accounts, any physical interaction, including an AI agent's information processing, would constitute observations sufficient to

resolve superposition. The human-specific effect reported here is therefore in tension with those frameworks as well as with the null hypothesis.

Each of the remaining seven nulls was as theoretically informative as the predicted effect, because by design the participants had no target to observe in those cases. The AI $\times$ observed $\times$ intention cell was null despite the AI extracting structured information from the same prerecorded data through the same browser interface. This implies that information extraction per se did not drive the effect.

4.2 The unobserved sensor

Although the human $\times$ unobserved $\times$ intention trend ($p = 0.074$, slope $\approx 29\%$ of the observed-channel magnitude) was not preregistered, its direction and magnitude mapped onto a specific quantitative prediction of the vN-W framework that, to our knowledge, has not been previously articulated. We develop that prediction here as exploratory, recognizing that fitting parameters to data to motivate a model is not a confirmatory test.

The idea is this: The two first-order diffraction spots are not independent channels but two angular slices of one photon field. When observation projects the photon state onto the $+1$ mode, Heisenberg back-action broadens that mode into a Gaussian of angular width σ ; the Gaussian tail reaches the unobserved -1 location with fractional amplitude $\exp(-\text{sep}^2/2\sigma^2)$ (with “sep” meaning separation), so the dimensionless ratio $\tilde{\sigma} = \sigma/\text{sep}$ alone determines the cross-channel coupling (see §S3). For $\tilde{\sigma} \rightarrow 0$ (a sharply localized measurement) the cross-channel ratio vanishes; for $\tilde{\sigma} \approx 0.7$ it is approximately 0.36. Conscious observation of a bright spot, on this reading, is not sharply localized in the way a fiber-coupled single-photon detector would be. Rather, the observer's effective mode of engagement with the optical field is spatially diffuse. Working backwards from the data, the empirical cross-channel ratios (intention-only 0.300; intention-relax 0.404) yield $\tilde{\sigma} = 0.64$ and 0.74, a back-action footprint roughly two-thirds of the spot separation, which is physically plausible with a measurement mode whose spatial footprint is comparable to the diffraction geometry itself.

Out-of-sample validation. Fitting $\tilde{\sigma}$ to the same data that motivated the model would be circular. We therefore fit $\tilde{\sigma}$ on the pre-preregistration window ($n = 817$), then held the value fixed (Int-Rel $\tilde{\sigma} = 0.706$), and predicted the post-preregistration cross-channel ratio without parameter

adjustment. The predicted unobserved Intention-Relax slope was -1.82×10^{-4} , and the actual slope was -2.83×10^{-4} . The bootstrap 95% confidence interval (CI) on the prediction error contained zero, and the test-window $\tilde{\sigma}$ (0.94) fell within the training-window bootstrap CI. With the caveat that these CIs are wide, the model trained on pre-preregistration data correctly predicted the direction, sign, and order of magnitude of the post-preregistration unobserved-channel behavior. Full CIs and the parallel intention-only prediction are reported in §S4.

Spatial-gradient prediction. The model generates a falsifiable prediction beyond the observed/unobserved comparison: the zeroth-order spot, at angular distance $\text{sep}/2$ from the first-order observed spot, should show approximately 75% of the first-order effect (from $\exp(-1/(8\tilde{\sigma}^2)) \approx 0.775$ at $\tilde{\sigma} \approx 0.7$), and second-order spots should show smaller but nonzero effects. Simultaneously monitoring these additional spots in future studies would provide a test of this broadened-detector-mode model.

A prior diffraction-based paradigm did record data from both first-order spots and the central spot [43]. The observed first-order spot showed a clear monotonic decline, the zeroth-order central spot showed a similar but smaller decline, and the unobserved first-order spot showed a reversal. The first-order observed and central-spot results match the model's Gaussian-gradient prediction. The unobserved-spot reversal differed from the small same-sign drift observed in the present study, but inexpensive diode lasers were used in that experiment and their intensity drift was at times comparable to the back-action signal itself; by comparison the present study used an ultra-low-noise laser. A same-sign decrease of the magnitude reported here ($\approx 30\text{--}40\%$ of the observed sensor) would lie below the per-session noise floor of the prior apparatus, and under sufficient residual drift it could appear as a sign flip in any single recording window. However, in that study the zeroth-order gradient evidence was unaffected by these noise considerations, possibly because the central-spot measure was a within-session simultaneous-recording comparison, providing support for the Gaussian model that is independent of the present study. Again, this model is speculative pending future replications.

4.3 The retrocausal design

The back-action argument is forward-causal. The observer measures a photon, back-action imparts momentum spread, and the spot dims. The experimental design used here, however,

was retrocausal [60–62]. Sensor data were recorded months before any participant observed them, and no experimenter ever observed the data. If the photons landed where they landed in June 2025 and the voltages were written to disk, what was left for a 2026 observer to collapse?

Recording a voltage on a hard drive is a classical operation, and decoherence theory holds that environmental interactions would destroy quantum coherence on femtosecond timescales for macroscopic objects. From that view, the retrocausal design cannot be explained by standard forward-causal back-action. If the effect observed here was real, then its explanation requires an extension of the standard framework. The crucial distinction then is between decoherence (the suppression of interference between branches) and actualization (the selection of a specific branch).

A treatment by Levin [46], working within the Gell-Mann–Hartle decoherent-histories formalism, makes this point mathematically. Decoherence alone, Levin notes, does not solve the “problem of outcome.” It explains why interference between alternatives is suppressed in macroscopic records, but not why one alternative rather than another is actualized. Within orthodox quantum mechanics the selection requires a von Neumann Process-1 collapse, which is associated with an agent's mental event. A recording that has decohered but has not been observed by a conscious agent therefore still consists of coexisting decoherent alternatives. The eventual conscious observation supplies the end condition that retroactively fixes which alternative is realized, generating the appearance of pre-observation correlations in retrospect.

Levin develops this account specifically to explain a “presentiment” effect [63,64], an experimental paradigm in which pre-stimulus physiological measures correlate with randomly selected future stimuli. Levin's prediction was that physiological data recorded in the absence of a subsequent conscious observer would show no prestimulus effect. The AI cohort in the present study is structurally analogous in that an identical apparatus, identical recordings, and an identical analysis pipeline were used, but with the conscious-observer condition replaced by an AI agent. Within the vN-W framework, an AI agent (at least the models available as of mid-2026) is not the kind of system expected to supply a Process-1 end condition, and the null AI result is thus compatible with Levin's prediction.

Two frameworks from established physics also provide footholds. Wheeler's delayed-choice principle, experimentally confirmed in multiple photonic implementations [65,66],

demonstrates that the choice of what to measure, made after a photon has transited an apparatus, can seemingly retroactively determine which mutually exclusive history the photon appears to have had. That said, it is worth mentioning that some argue that the delayed choice quantum eraser neither erases nor delays [67], while others maintain that while “there is no physical effect of the present or of the future on the past, we must nevertheless consider that the observer’s past is sometimes not entirely determined and that it becomes determined only when certain measurements are done later” [68].

Still, the present design tests a stronger and more controversial extension of Wheeler’s idea, namely that a recorded but consciously unobserved optical history may remain open to later selection effects. Aharonov’s two-state vector formalism (TSVF) is thus more directly applicable [61,69]. With TSVF a quantum system at any moment is characterized simultaneously by a state propagating forward from preparation and a state propagating backward from its future measurement. The future observation contributes a backward-evolving state that intersects the earlier recording, and the back-action mechanism (involving momentum broadening, resulting in spot dimming) operates in both temporal directions through the same Gaussian form.

We do not claim these frameworks fully explain the present results. The applicability of delayed-choice and TSVF reasoning to macroscopic classical data records is not well established, and the claim that stored data retains residual quantum openness is, if confirmed empirically, a significant challenge to standard decoherence theory. We mention them to show that a retrocausal effect, if real, has a theoretical basis in existing physics rather than requiring entirely new physics. The retrocausal design also delivers a control that no forward-causal version can in that it completely decouples classical environmental influences between participant and apparatus. The AI cohort tightened this control even further. If decoherence neutralized the recorded data, then the retrocausal design would be physically equivalent to a forward-causal one, and any apparatus or pipeline artifact would appear in the AI sessions as well as the human sessions. But given that it did not, the human–AI divergence was not merely a control against analytical artifacts but also a control against the alternative interpretation of the retrocausal design itself.

4.4 Addressing the classical-beam objection

Another possible objection is that a milliwatt continuous-wave laser, producing $\sim 10^{15}$ – 10^{16} photons per second, behaves as a classical electromagnetic field, leaving nothing for a quantum collapse mechanism to act upon. Several considerations argue against this objection. First, the classical field description is an approximation that breaks down when measurement back-action is examined at the level of individual detection events. Aharonov, Albert, and Vaidman demonstrated that weak-coupling measurement yields physically anomalous results [70], confirmed experimentally with a continuous beam by Ritchie et al [71]. High photon flux provides the statistical ensemble over which weak values are defined, and human observation in the present case constitutes an ultra-weak measurement in precisely this sense.

Second, the objection would apply equally to similar interferometer experiments cited in the Introduction, all but one of which used continuous laser beams. A reader who finds the objection decisive for the present study must explain why the existing body of evidence is not similarly disqualified. Third, within quantum optics a coherent laser state remains in a superposition of Fock states regardless of intensity, and the diffraction pattern is produced by individual photons in a superposition. And finally, under the retrocausal design the question is not whether a conscious observer can deflect a photon in real time but whether a decoherent-but-unobserved historical record has a definite outcome assigned, a conceptually distinct claim that is orthogonal to photon flux at the time of recording.

The power-level dependence reported in section 3.4 bears directly on this objection. The two first-order diffraction channels operated at a $\sim 4:1$ power ratio. A classical signal-to-noise argument would predict that the higher-power channel should show the stronger effect because it provides more photons per measurement interval and thus a better-defined signal. However, the opposite was found, with the effect-to-noise ratio $\sim 1.4\times$ larger on the low-power channel. This inverse relationship between photon flux and effect magnitude is expected if consciousness interacts with individual quantum measurement events rather than with the aggregate intensity of the beam. Under the vN-W framework, each photon collapse event is an elementary act of actualization; when photon flux is lower, each individual event contributes proportionally more to the measurable aggregate, and the fractional influence of the observer is

therefore larger. High photon flux, on this account, would not amplify the collapse effect; it would instead dilute it.

4.5 Optional stopping as an alternative explanation

One possible concern is that the engagement criterion, which required participants to correctly identify at least 9 of 10 words, might have introduced an optional stopping bias. If participants who sensed the experiment was producing desirable results were more likely to remain engaged and complete all 10 epochs, then that criterion might have disproportionately selected sessions in which the effect happened to be stronger, thus inflating the results.

Several lines of evidence argue against this interpretation. First, the effect was not specific to the 9/10 threshold. A sweep across all engagement levels from 1/10 to 10/10 correct showed that the observed-sensor slope difference was statistically significant at every threshold, with Cohen's d stable between -0.027 and -0.037 throughout (see Figure 7). That is, the effect did not suddenly emerge at the preregistered cutoff; it was present even in sessions where participants identified as few as one word correctly. If the 9/10 criterion were generating a spurious effect through selective inclusion, one would expect the signal to appear at that threshold rather than be uniformly present below it.

Second, inspection of the raw session data showed that sessions with fewer than 10 recorded responses represented genuine early exits rather than silent presence. Of 71 sessions with only one response, all 71 had no timestamps beyond the first epoch; this confirmed that participants quit the task rather than remaining present without responding. These sessions are fewer than 7% of the qualifying pool, and their effect sizes are noisy and centered near zero. This is more consistent with small samples from a null distribution rather than evidence of a suppressed signal.

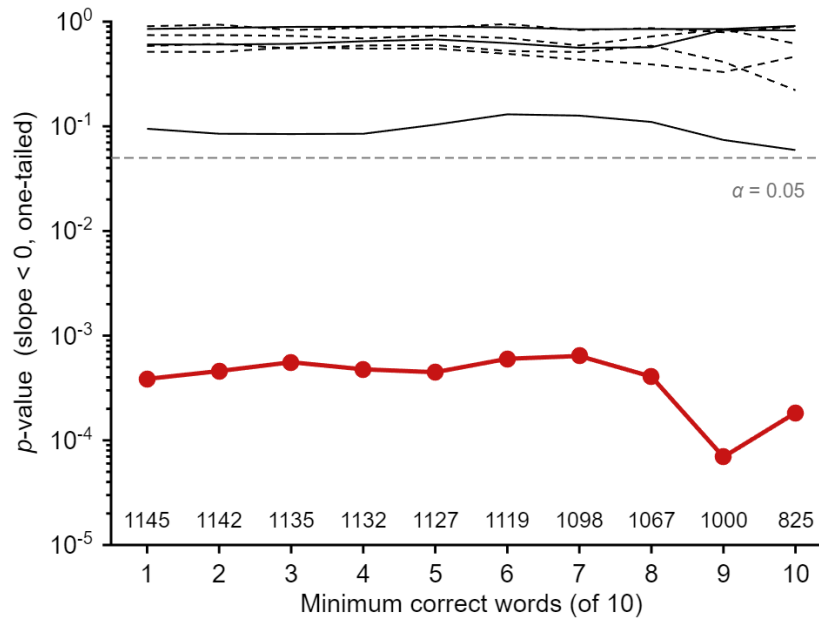


Figure 7. The red line (filled circles) shows the one-tailed p-value for the Human $\times$ Observed $\times$ Intention ensemble slope. The three solid black lines show the three Human null cells (Observed Relax; Unobserved Intention; Unobserved Relax). The four dashed black lines show the four AI null cells (the same four combinations for AI agents). Numbers along the bottom axis indicate total sessions at each threshold. The dashed gray horizontal line marks $\alpha = 0.05$. All seven null curves remain above 0.05 across every threshold; the predicted cell remains below $p < 10^{-3}$ throughout.

A possible counterargument runs as follows: A participant who only answered 1 of 10 words correctly was still assigned to a pre-recorded data segment, so that segment's “signal,” whatever it was, already existed. The engagement threshold (e.g., 1 or 10 correct responses) did not change the recording; it only determined how carefully participants were curated and linked to which segments. Thus, it was not surprising that even low participant engagement (e.g., 1 correct response) showed an effect, because the effect already resided in the pre-recorded data and not in what participants did in real time. This suggests that the results may have been due to an artifact; e.g. perhaps the data were inherently biased toward negative slopes in the intention segments, and thus the results had nothing to do with a consciousness-mediated effect.

However, that interpretation is ruled out by the controls. The unobserved sensor, which was switched in each successive session, used the same source of prerecorded data and the same analytical pipeline. If the data were inherently biased, both sensors would have shown it. But

only the human $\times$ intention $\times$ observed condition did. Likewise, the AI agents used the same recordings, same linking procedure, and same pipeline, but all of those results were also null. Thus, the pre-recorded data were not intrinsically biased, and something specific about the human-session pairings drove the result.

4.6 Limitations and future directions

A key limitation in this study was that the effect size, Cohen's $d \approx -0.03$, was minuscule. At this scale, to distinguish signal from artifact the burden falls primarily on internal controls.

Applying the same analytical pipeline to all 8 cells in the observer $\times$ sensor $\times$ task matrix produced significant results only in the one predicted cell. A systematic bias of any classical origin (sensor drift, pipeline artifact, session-timing confound, etc.) would have needed to selectively affect only that one cell while leaving all others unaffected. We are not aware of any plausible mechanism that could satisfy this constraint.

Another important limitation was that the preregistration was filed after some pilot data were collected. The five-criterion replication analysis showed that the effect persisted past the early-window data and was consistent with the pilot results (also see §S1), but a fully prospective replication from the first session would be an essential component of a replication study.

The broadened-detector-mode out-of-sample test (§S4) confirmed the direction and order of magnitude for the human $\times$ intention $\times$ unobserved sensor outcome, but not a precise measure of $\tilde{\sigma}$. A more stringent future test would simultaneously instrument the zeroth-order and second-order spots in addition to the two first-order spots to map the full angular profile of the back-action gradient predicted by the model.

5 Conclusion

Across 1,000 qualifying human sessions, the preregistered prediction was satisfied. During intention epochs, the sensor data that later drove the participant's display showed a small magnitude but statistically reliable negative slope relative to relax epochs, whereas the unobserved sensor and the AI-agent sessions did not. The AI condition was especially important because it used the same prerecorded optical data, browser interface, and analysis pipeline, but without a human observer intentionally engaged with the display.

The effect size was minuscule, the preregistration was filed after some human data had already been collected, and the broadened-mode account of the unobserved-channel trend remains speculative. For these reasons, the results of this test should be regarded as promising rather than as an established demonstration. Still, the pattern is specific, internally controlled, and several mundane explanations were ruled out by design and by the AI null result in a way that apparatus or analytical artifacts would not readily explain. A prospective replication with simultaneous monitoring is the next step.

Author Contributions

DR designed the experimental protocol, the web and mobile applications, created the optical apparatus and recorded the sensor data, analyzed the results, and wrote the first draft of this manuscript; RC contributed the design and code for the AI agent aspect of the test and edited the draft.

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Data Availability

The analysis code, preprocessed sensor data, and session records supporting this study are publicly available at Zenodo: Radin, D., & Cline, R. (2026). Noetic Adventure — Analysis code and data (Version 1.0.3). Zenodo. <https://doi.org/10.5281/zenodo.20099018>. An interactive notebook that runs in the browser without installation is available at https://colab.research.google.com/github/deanradin/noetic-adventure/blob/main/noetic_adventure_analysis.ipynb

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Claude Code (Anthropic, Opus 4.6 and 4.7) was used to assist in preparing analytical code and programming the AI agents. All results were checked and confirmed by the authors.

Competing Interests

The second author was one of the funders of this test and contributed to the development of the AI agent portion of the test. All other aspects of the test were the responsibilities of the first author. A provisional patent application was filed on the optical apparatus and signal-processing methods described herein, motivated solely by the need to protect intellectual property arising from fundamental research. The patent pertains to potential engineering applications of retrocausal measurement techniques and does not influence the scientific claims, data analysis, or preregistration of this study.

Ethics Approval

The protocol and informed consent language for the reported experiment was approved by the Institute of Noetic Sciences Institutional Review Board (RADD_226_01).

Supplementary Materials

§S1. Effect consistency pre- to post-preregistration

Figure S1.1 plots the cumulative Stouffer Z separately for human and AI cohorts on observed and unobserved sensors as a function of session count, and Figure S1.2 plots the same data as one-tailed odds against chance on a log scale.

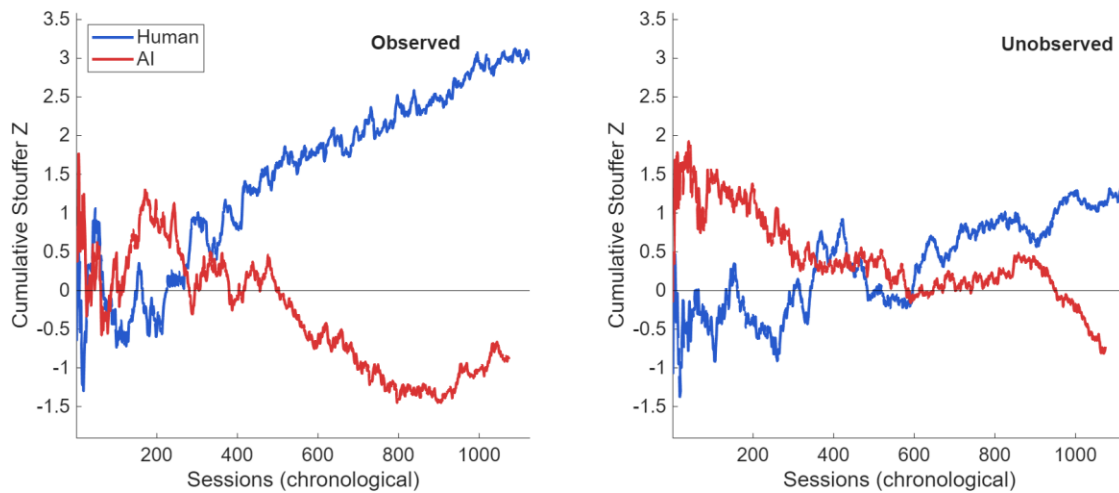


Figure S1.1. Cumulative Stouffer Z of the difference between intention and relax epochs across all human and AI sessions in the observed and unobserved sensors.

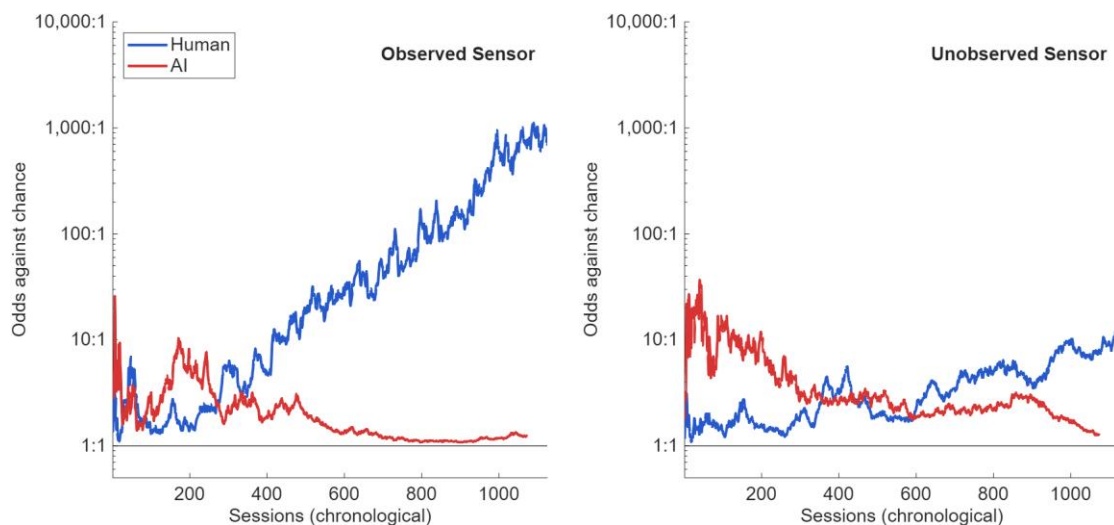


Figure S1.2. Cumulative one-tailed odds against chance of the difference between intention and relax epochs across all human and AI sessions in the observed and unobserved sensors.

§S2. Lag analysis

The preregistered 5-second cognitive-switching delay was checked by sweeping across lags from 0 to 10 seconds, based on 10,000 permutations of observed and unobserved intention versus relax slopes. As shown in Figure S2, this exercise identified 4.5 s as the optimal delay.

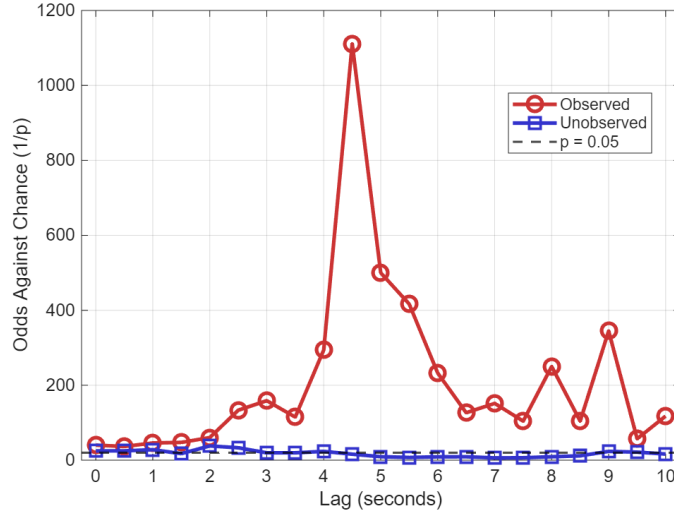


Figure S2. Results comparing intention versus relax slopes for observed and unobserved sensors in terms of odds against chance, with lags ranging from 0 s to 10 s.

§S3 Derivation of the cross-channel coupling ratio

The two first-order diffraction spots arise from photons that pick up a transverse momentum kick of $\pm\hbar(2\pi/d)$ at the grating, deflecting to angular locations $\pm\lambda/d$ ($= \pm\text{sep}/2$) from the optical axis, where $\lambda = 635$ nm, d = grating period, and sep = total angular separation between +1 and -1 first-order spots.

The two spots are not independent detection channels, but two angular slices of a single photon field. The Heisenberg uncertainty principle, applied to the measurement interaction, requires that any localization of a photon's angular position induce a random momentum kick (back-action) that is proportional to the precision of the localization. In the present geometry, this broadens the +1 spot from a near-delta-function into a Gaussian intensity profile of angular width σ :

$$I(\theta) \propto \exp(-(\theta - \text{sep}/2)^2 / 2\sigma^2).$$

On the vN-W reading, this broadening is not a redistribution of conserved photon flux but an angular profile of the back-action itself. The entire diffraction pattern is attenuated, with the attenuation strongest at the attended location and falling off as a Gaussian. We define the attenuation profile $A(\theta) = \exp(-(\theta - \text{sep}/2)^2 / 2\sigma^2)$, centered on the +1 spot at $\theta = \text{sep}/2$, so that $A(\text{sep}/2) = 1$ (maximum attenuation at the attended location) and $A(\theta) < 1$ elsewhere. The relative attenuation at angular position θ relative to the attended spot is therefore $A(\theta)/A(\text{sep}/2) = A(\theta)$, giving the cross-channel ratio:

$$A(-\text{sep}/2) / A(\text{sep}/2) = \exp(-\text{sep}^2 / 2\sigma^2).$$

Note that because $A(\text{sep}/2) = 1$ by construction, the ratio $A(\theta)/A(\text{sep}/2)$ is just $A(\theta)$, so one could simply say that the relative attenuation at position θ is $A(\theta)$ without the denominator. Now, defining $\tilde{\sigma} \equiv \sigma / \text{sep}$, the cross-channel coupling ratio depends only on $\tilde{\sigma}$:

$$R(\tilde{\sigma}) = \exp(-1 / 2\tilde{\sigma}^2).$$

For $\tilde{\sigma} \rightarrow 0$ (sharply localized measurement, fiber-coupled detector limit) $R \rightarrow 0$ and the unobserved spot is untouched. For $\tilde{\sigma} \approx 0.7$ (a measurement mode comparable in scale to the diffraction geometry itself) $R \approx 0.36$. Inverting $R(\tilde{\sigma})$ gives $\tilde{\sigma}$ from any empirical cross-channel ratio: $\tilde{\sigma} = \sqrt{-1 / (2 \ln R)}$.

For the central (zeroth-order) spot at angular distance $\text{sep}/2$ from the +1 location, the equivalent ratio is $\exp(-1 / (8\tilde{\sigma}^2))$, giving $R_{\text{zero}} \approx 0.775$ at $\tilde{\sigma} \approx 0.7$, i.e., a same-sign decline of approximately 75% of the first-order effect.

§S4. Out-of-sample validation

For the pre-preregistration window (training data): $N = 817$ qualifying sessions, 8,170 blocks. For the post-preregistration window (test): $N = 308$ sessions, 3,080 blocks (extended beyond the 1,000-session preregistered stopping rule and thus exploratory).

Training-window $\tilde{\sigma}$ values (point estimate; 95% bootstrap CI from 5,000 iterations):

- Intention-only: $\tilde{\sigma} = 0.64$ [0.471, 1.073]

- Intention-relax: $\tilde{\sigma} = 0.71$ [0.519, 1.154]

Held-fixed predictions (parameter-free): $\text{unobserved_test} = \text{observed_test} \times \exp(-1 / (2\tilde{\sigma}^2))$.

Intention-only slopes:

- Predicted unobserved: -1.21×10^{-4}
- Actual unobserved: -1.62×10^{-4}
- Prediction error 95% CI: $[-4.3 \times 10^{-4}, +4.3 \times 10^{-4}]$ (contains zero)
- Test-window $\tilde{\sigma}$: 0.74 (within training-window CI)

Intention-relax slope difference:

- Predicted unobserved: -1.82×10^{-4}
- Actual unobserved: -2.83×10^{-4}
- Prediction error 95% CI: $[-6.5 \times 10^{-4}, +5.6 \times 10^{-4}]$ (contains zero)
- Test-window $\tilde{\sigma}$: 0.94 (within training-window CI)

Note: The wide confidence intervals on $\tilde{\sigma}$ are a consequence of estimating a Gaussian coupling coefficient from noisy slope differences at realistic sample sizes. The out-of-sample test should therefore be read as confirming the direction, sign, and order of magnitude of the cross-channel coupling, and not as providing a precise determination of $\tilde{\sigma}$. However, within those limits the model trained on pre-preregistration data predicts the post-preregistration unobserved-channel behavior without parameter adjustment.

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